Pranjal Singh

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Professional Summary

High-impact **Full Stack Developer & Systems Engineer** with deep expertise in architecting high-concurrency microservices, reactive frontend web platforms, and low-latency distributed systems. Proficient across modern TypeScript, React 19, Next.js, Golang, Python, and Node.js with production experience designing REST/WebSocket streaming protocols, Zero-Trust network gateways, and automated AST developer tooling. Proven track record leading technical teams, optimizing database engines, and shipping scalable cloud products with 99.9% uptime.

Experience

- **Stealthera Innovations Pvt. Ltd.** Remote
  AI Technical Lead *Jun 2026 – Present*
  - Spearheaded end-to-end full-stack architecture and engineering strategy for real-time speech intelligence pipelines, coordinating cross-functional teams to deliver streaming voice applications with sub-50ms processing latency.
  - Engineered distributed microservice backends and audio DSP pipelines using Faster-Whisper, WebSockets, and INT8 ONNX quantization; slashed model inference latency by **42%** and reduced server memory footprint by **35%**.
  - Designed reactive telemetry dashboards with continuous audio stream visualizations, integrating automated evaluation suites (Evals) to benchmark Word Error Rate (WER) and p95 latencies across high-noise enterprise environments.
  - Directed sprint planning, code reviews, and CI/CD automated regression pipelines, eliminating edge-case failure modes and cutting production drift incidents by **35%**.

Flagship Full Stack & Systems Projects

- **PocketPaisa (Tijori)** | *React 19, TypeScript, Node.js, Express, MongoDB Atlas, Stripe API, Recharts, Zod* | Live Demo | GitHub 2026
  - Architected a full-stack personal finance web application featuring automated 50-30-20 budget envelopes, digital goal vaults ("Gullaks") with atomic deposit/withdrawal workflows, and 1-tap rapid expense logging (i2s execution).
  - Engineered a real-time "Can I Afford This?" impulse evaluation engine calculating cashflow burn velocity and dynamically computing month-end daily allowance projections; integrated MongoDB aggregations to detect micro-leakage drains.
  - Implemented dual-mode payment architecture with Stripe Checkout API and a frictionless 1-click sandbox gateway, complete with secure JWT authorization, Google OAuth2, and automated Gmail SMTP security alert notifications.
- **Aegis Edge Gateway** | *Golang, Python, React 19, TypeScript, Nginx, WebSocket, Docker* | Live Demo | GitHub 2026
  - Engineered a production-ready Zero-Trust reverse proxy in Go featuring Attribute-Based Access Control (ABAC), token-bucket rate limiting, dynamic upstream routing, and automatic JWT security header verification with sub-3ms p95 latency.
  - Integrated a Python Threat Intelligence engine calculating real-time Shannon Entropy and regex heuristic scores to inspect payloads, intercepting obfuscated XSS, SQLi, and DOM injection anomalies before upstream transmission.
  - Constructed a real-time SOC observability dashboard in React 19 streaming live packet forensic traces via low-latency WebSockets; containerized complete multi-service architecture with Nginx edge routing in Docker.
- **HELIOS Workbench** | *Next.js 14, TypeScript, Ollama (Qwen 2.5), Turborepo, OpenAPI AST* | Live Demo | GitHub 2026
  - Built an enterprise developer workbench compiling OpenAPI 2.0/3.0/3.1 specifications into a canonical in-memory Abstract Syntax Tree (AST), powering offline local RAG AI reasoning with **0%** hallucination.
  - Automated compliance scoring against PCI-DSS 4.0, HIPAA, and OWASP Top 10 standards via AST visitor passes; built breaking-change tree-diffing engines and automated SDK generators across 7 programming languages.

Technical Skills

Languages: TypeScript, JavaScript (ES6+), Go (Golang), Python, Rust, SQL, C/C++, HTML5/CSS3
Frontend: React 19, Next.js 14, Tailwind CSS, Redux Toolkit, Zustand, Vite, Recharts, WebSockets, Responsive UI/UX
Backend & Systems: Node.js, Express.js, Go `net/http`, FastAPI, RESTful APIs, Microservices, Reverse Proxies, Nginx, Linux/POSIX
Databases & DevOps: MongoDB Atlas, PostgreSQL, Redis, SQLite, Docker, Turborepo, Git/GitHub, CI/CD, Render, Vercel

Education

- **Indian Institute of Information Technology, Ranchi (IIIT Ranchi)** Ranchi, India
  Bachelor of Technology in Electronics and Communication Engineering (CGPA: 8.27 / 10.0) *2024 – 2028*

Honors & Achievements

- **Smart India Hackathon (SIH)**: Qualified university rounds to represent IIIT Ranchi at the national-level competition, architecting high-impact distributed solutions under 36-hour sprint deadlines.
- **Algorithmic Problem Solving**: Solved **200+** Data Structures and Algorithms problems across LeetCode and CodeChef; achieved a peak contest rating of **1553** on CodeChef.